



Introduction

- Bayesian calibration is suited for health decision models because they estimate the posterior joint and marginal distributions of model input parameters.
- It can be computationally expensive requiring evaluating the model thousands, if not millions of times.
- Emulator-based calibration can reduce this burden.
- We used Bayesian calibration with artificial neural networks (BayCANN), an emulator-based algorithm, to calibrate CISNET's SimCRC, MISCAN-Colon, and CRCSPIN natural history colorectal cancer (CRC) simulation models and to validate the model predicted outcomes to calibration targets.

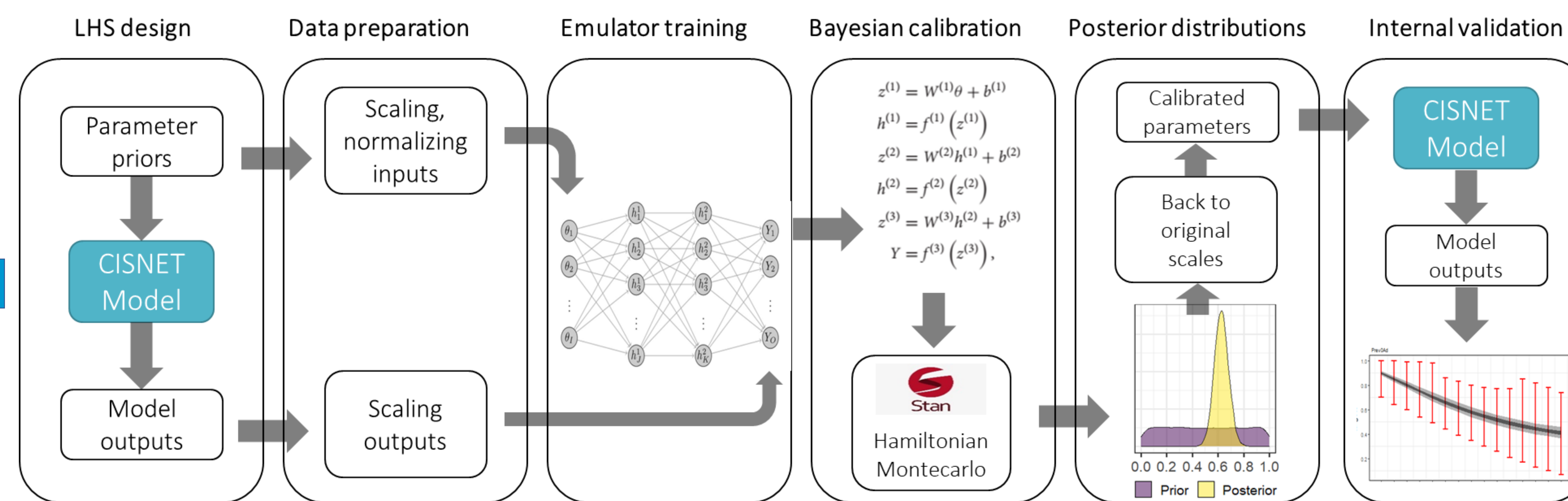
Methods

BayCANN main steps on each CISNET model (Fig 1).

- Sample parameters using a Latin hypercube sampling (LHS) design.
- Generate model-predicted outputs for each parameter set from the LHS.
- Verify that model outputs covered target uncertainty.
- Transform model inputs and outputs to train the ANNs.
- Optimize hyperparameters (hidden layers, hidden nodes, activation function, batch, optimizer) and ANN structures by minimizing the mean square error (MSE).
- Code the optimal ANN for each CISNET model in Stan.
- Use Hamiltonian Monte Carlo methods to estimate the posterior distribution.
- Transform inputs back to original scale and propagate uncertainty on model predicted outputs.
- Internal validation by comparing model-predicted outputs from BayCANN against the calibration targets.

Emulators can efficiently calibrate complex microsimulation models and reduce the computational and time burden for calibration.

Fig 1. Implementation steps on CRC models



Results

Table 1. Training and calibration time.

Model	Training time	Calibration time
SimCRC 30 parameters 110 targets	39 min	470 min
MISCAN 37 parameters 84 targets	26 min	214 min
CRC SPIN 22 parameters 41 targets	10 min	14 min

Fig 2. Emulators' prediction performance. Comparison of model outputs with emulator-predicted outputs.

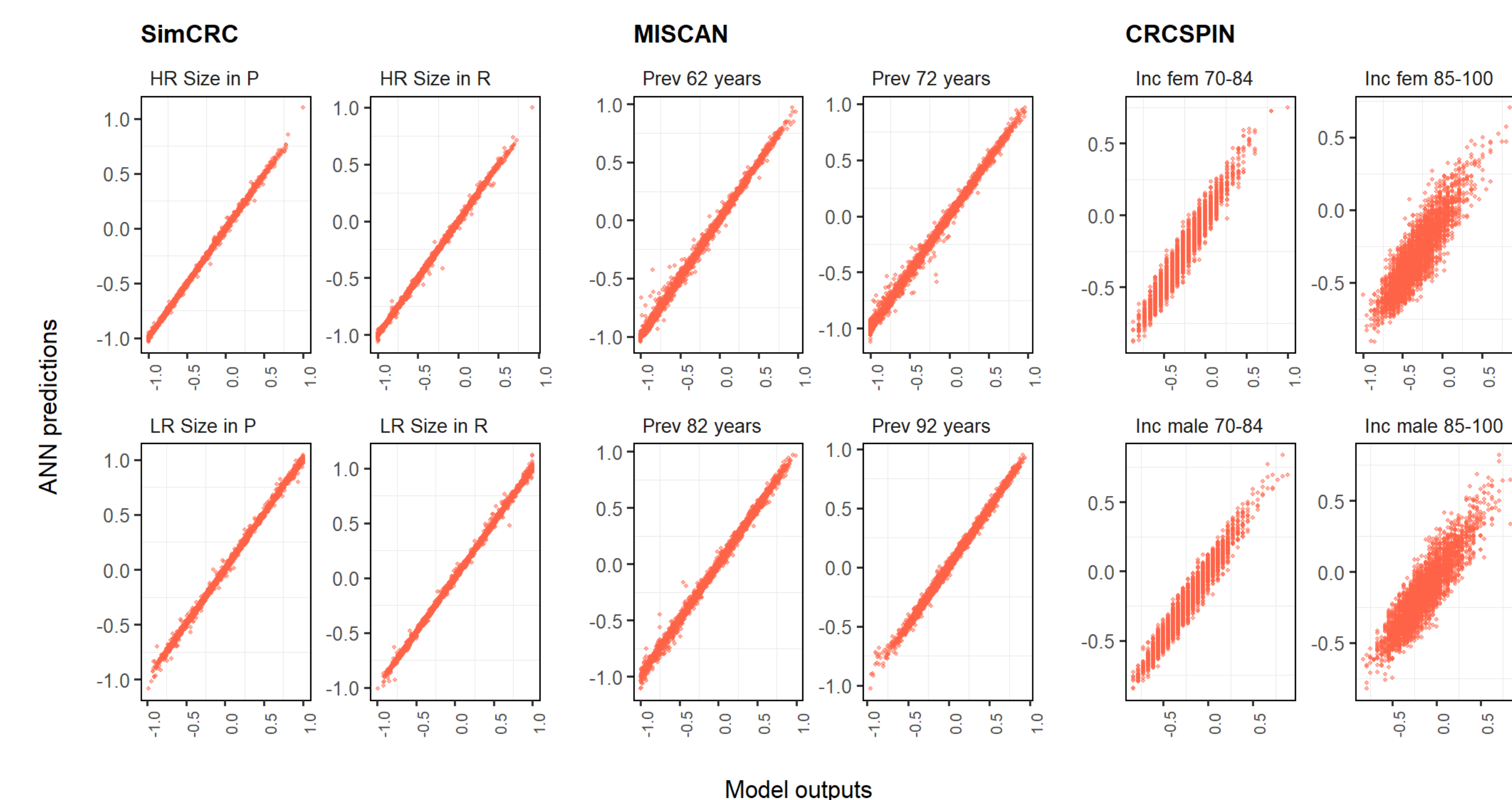


Fig 3. Calibrated posterior distributions. Prior and Marginal posterior distributions of calibrated parameters.

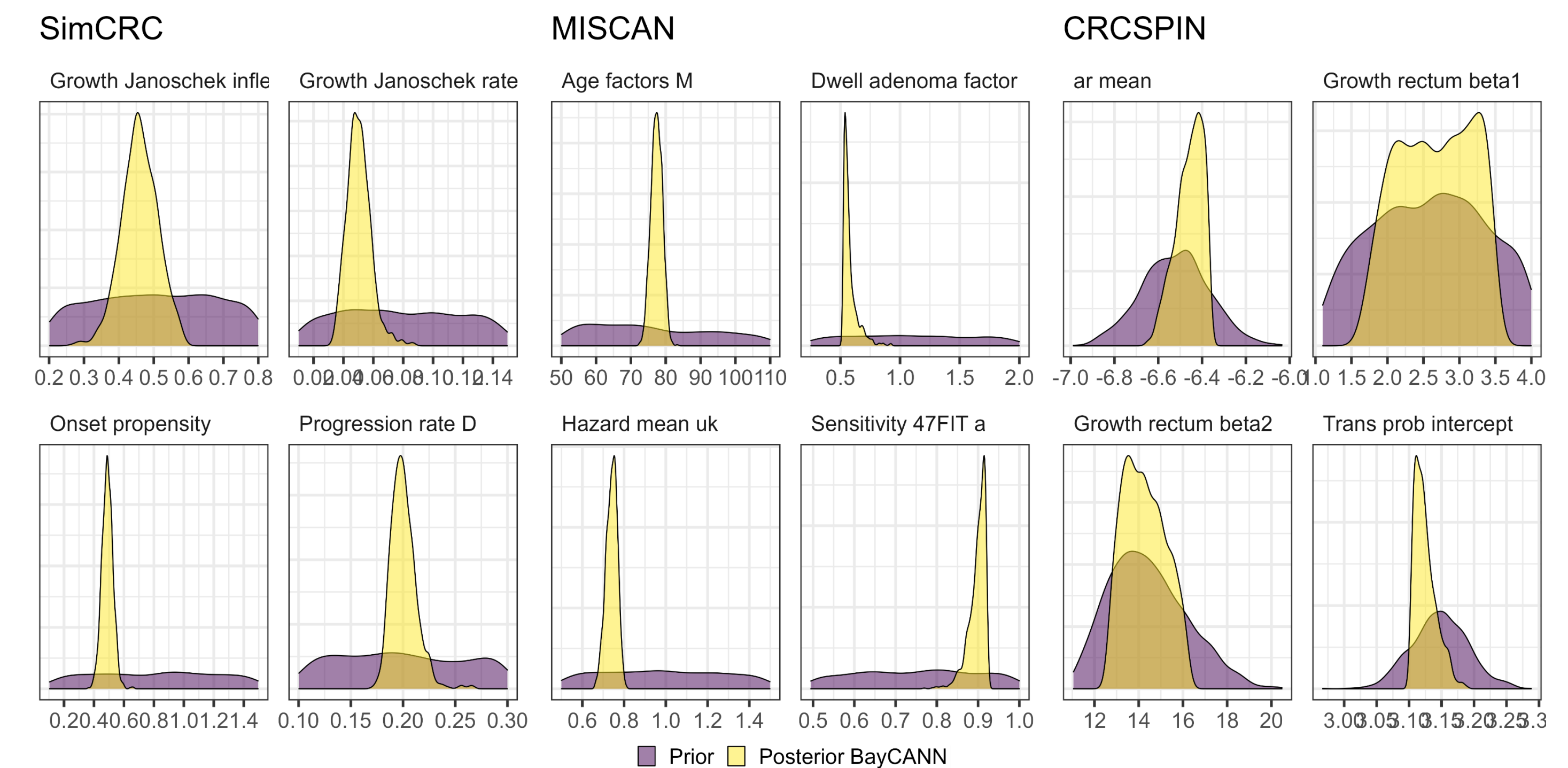
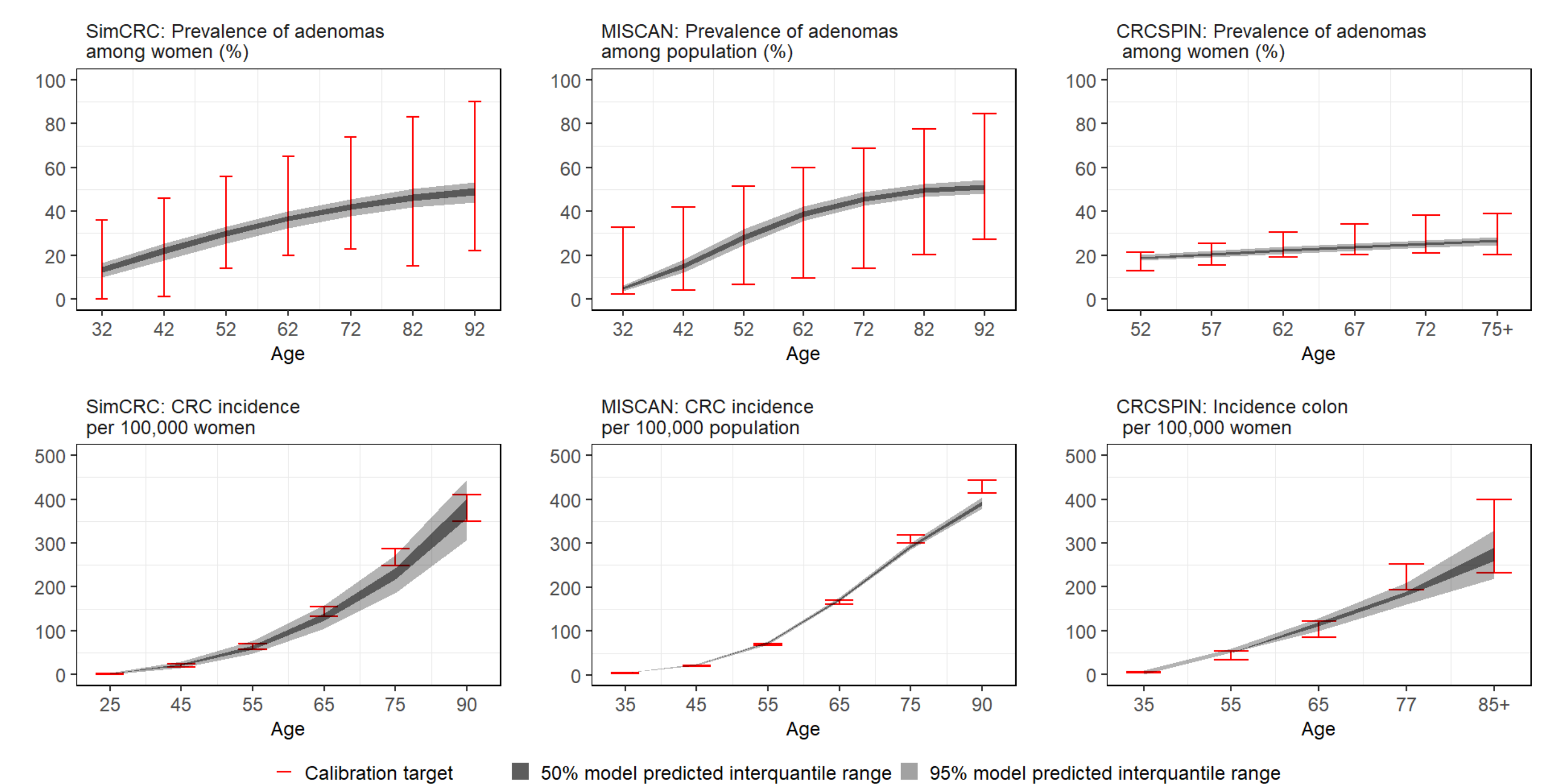


Fig 4. Internal validation. Validation for prevalence and incidence targets of SimCRC, MISCAN-Colon and CRC-SPIN.



Conclusions

- BayCANN only uses a dataset of model inputs and outputs for each CISNET model to obtain joint posterior distributions.
- ANNs performed well as metamodels of the CISNET models.
- We can implement BayCANN without coding the original CISNET models in a probabilistic programming language.
- The emulators required less than one hour to train and less than eight hours for parameter calibration.
- BayCANN can efficiently calibrate complex simulation models like the CISNET CRC microsimulation models.